

Sony Neural Network Console Desktopへの 移植と実行結果

宮野悟, 中林潤, 木下淳博, 須藤毅顕 「Pythonで体感！
医療とAIはじめの一步」 羊土社 [978-4758124188]

第5章の2つのコードを移植し、実行した結果

2025/01/15 H.Nishiyama

Main ✕ +

100% Action

Input

SquaredError

HuberLoss

AbsoluteError

EpsilonInsensitiveLoss

BinaryCrossEntropy

SigmoidCrossEntropy

CategoricalCrossEntropy

SoftmaxCrossEntropy

KLMultinomial

Parameter

WorkingMe

Affine

Convo

DepthwiseCo

Deconvolution

DepthwiseDeco

Diagram illustrating the architecture of a neural network for CIFAR-10, showing the flow of data and the number of parameters (w) and bias (b) for each layer.

- Input:** Dataset : x, 1,024,1024
- Interpolate:** OutputSize : 64,64, 1,64,64
- Reshape:** 4096
- Affine:** w, b, 512
- ReLU:** 512
- Affine_2:** w, b, 256
- ReLU_2:** 256
- Affine_3:** w, b, 128
- ReLU_3:** 128
- Affine_4:** w, b, 1
- Sigmoid:** 1
- BinaryCrossEntropy:** T, Dataset : y, 1

Run



Profile



Train



Structure Search



Evaluate

LOCAL

● HOST GPU

x1

○ HOST CPU

x1

Main



Network Statistics

Output	1,058,563
CostParameter	2,262,017
CostAdd	898
CostMultiply	0
CostMultiplyAdd	2,261,120
CostDivision	1
CostExp	1
CostIf	896

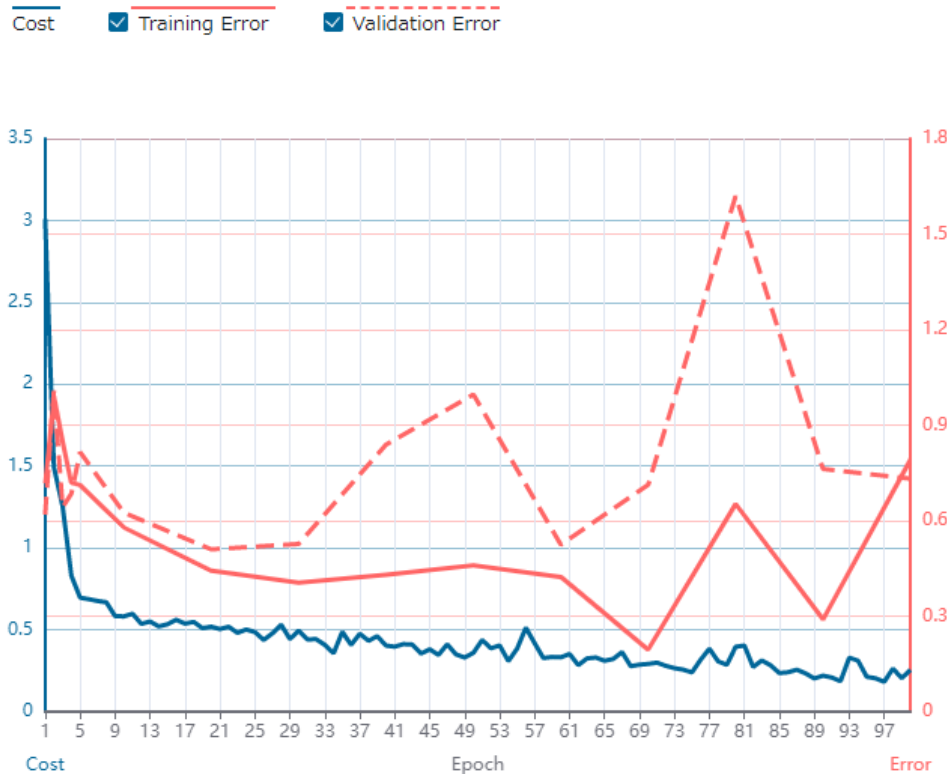
...

Epoch 100/100

Run

...

☐ Comparison Pareto
Optimal



```

nvidia command line interface (version: 1.9.0, Build: 240523022252, Callback: 10.2.0)
2025-01-15 19:10:33,719 [worker]: [INFO]: results handling...
2025-01-15 19:10:36,827 [worker]: [INFO]: create result_train_best.nnp
2025-01-15 19:10:36,842 [worker]: [INFO]: create result_train_last.nnp
2025-01-15 19:10:36,864 [worker]: [INFO]: create result.nnb
2025-01-15 19:10:38,958 [worker]: [INFO]: create result.onnx
2025-01-15 19:10:45,309 [worker]: [INFO]: create result.pb
2025-01-15 19:10:50,661 [worker]: [INFO]: worker done

```

x 1

Main



Network Statistics

Output	1,058,563
CostParameter	2,262,017
CostAdd	898
CostMultiply	0
CostMultiplyAdd	2,261,120
CostDivision	1
CostExp	1
CostIf	896

$y - y'$ ↕

Accuracy	0.7391		
Avg.Precision	0.7466		
Avg.Recall	0.7391		
Avg.F-Measures	0.7371		
	Recall	$y'=0$	$y'=1$
Precision		0.7895	0.7037
F-Measures		0.7143	0.76
$y:\text{label}=0$	0.6522	15	8
$y:\text{label}=1$	0.8261	4	19

Index	x:input	y'
1	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\covid.jpg  全体的に 白い肺	0.94849753
2	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\NORMAL.jpg  全体的に 黒い肺	0.23612474
3	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\test-01.jpg  全体的に 白い建物	0.5876318
4	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\test-02.jpg  全体的に 黒い建物	0.1747731

- Components
- Search
- IO

Input
- Loss

SquaredError

HuberLoss

AbsoluteError

EpsilonInsensitiveLoss

BinaryCrossEntropy

SigmoidCrossEntropy

CategoricalCrossEntropy

SoftmaxCrossEntropy

KLMultinomial
- Parameter

Parameter

WorkingMemory
- Basic

Affine

Convolution

DepthwiseConvolution

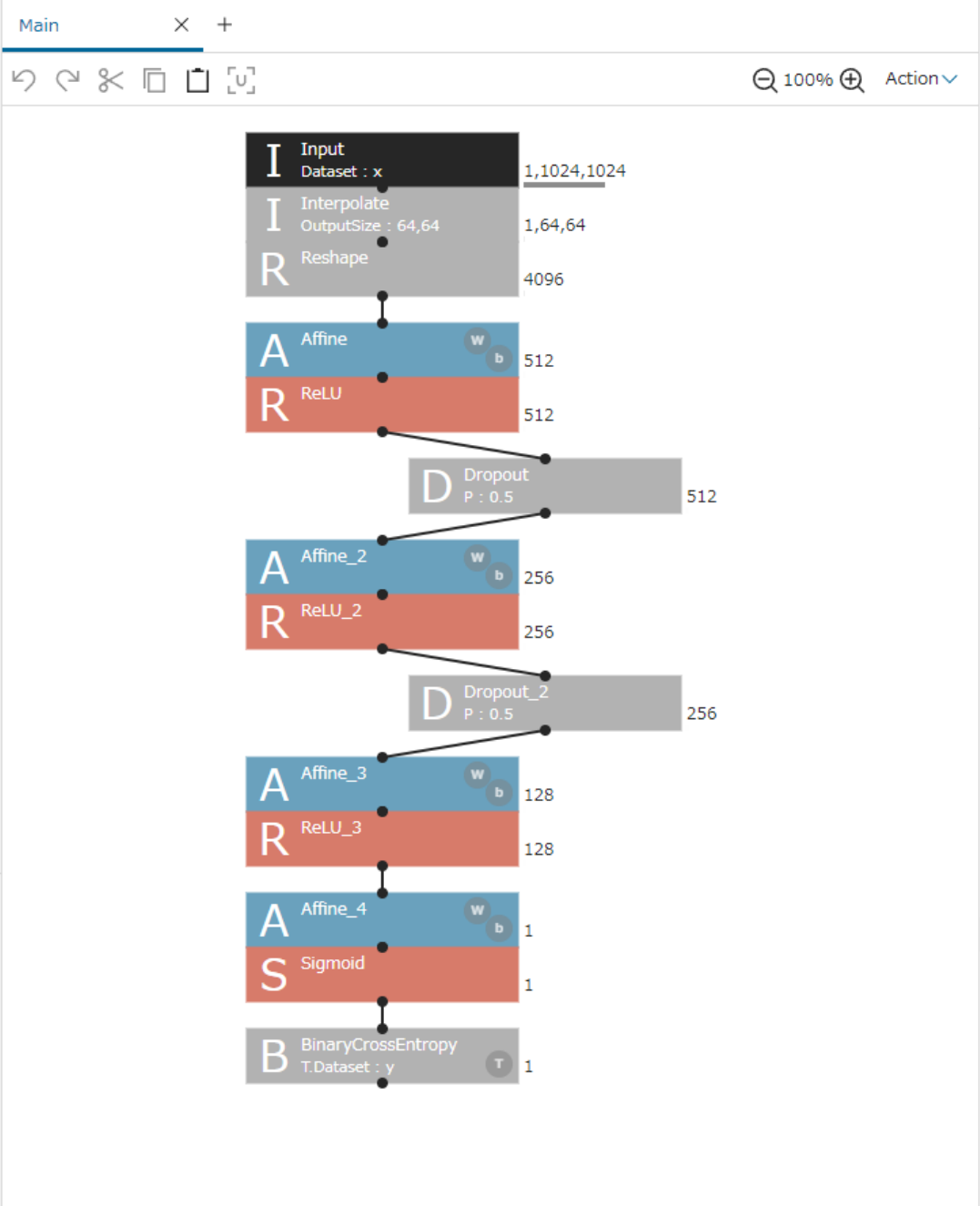
Deconvolution

DepthwiseDeconvolution

Embed
- Pooling

MaxPooling

Layer Property



Controller

Run

🕒 Profile

🖱️ Train ☐ Structure Search

✅ Evaluate

LOCAL

☒ HOST GPU x 1

☐ HOST CPU x 1

Overview

Main

Network Statistics

Output	1,059,331
CostParameter	2,262,017
CostAdd	898
CostMultiply	768
CostMultiplyAdd	2,261,120
CostDivision	1
CostExp	1
CostIf	896

CostIf	896
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$y - y'$

Accuracy	0.7391		
Avg.Precision	0.741		
Avg.Recall	0.7391		
Avg.F-Measures	0.7386		
	Recall	$y'=0$	$y'=1$
Precision		0.7619	0.72
F-Measures		0.7273	0.75
$y:\text{label}=0$	0.6957	16	7
$y:\text{label}=1$	0.7826	5	18

Index	x:input	y'
1	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\covid.jpg  全体的に 白い肺	0.99324256
2	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\NORMAL.jpg  全体的に 黒い肺	0.150386
3	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\test-01.jpg  全体的に 白い建物	0.72543395
4	C:\Users\HN\nncd_bucket\datasets\AI_Yodosha_test\data\test-02.jpg  全体的に 黒い建物	0.17743343